

Problem Set 1

1) Fun with estimating:

In humans, oxygen and carbon dioxide are exchanged in the blood within many small sacs called alveoli in the lungs. Alveoli provide a large surface area for gas exchange. Recent careful measurements show that the total number of alveoli in a typical pair of lungs is about 480×10^6 and that the average volume of a single alveolus is $4.2 \times 10^6 \mu\text{m}^3$. ($\mu\text{m} = 10^{-6}\text{m}$)

- a) Estimate the total volume of the gas-exchanging region of the lungs?
- b) If we assume that alveoli are spherical, what is the diameter of a typical alveolus?

2) Linear deceleration problem.

A neutron travels at a speed of $1.4 \times 10^7 \text{ m/s}$. Nuclear forces are of very short range, being essentially zero outside a nucleus but very strong inside. If the neutron is captured and brought to rest by a nucleus whose diameter is $1.0 \times 10^{-14} \text{ m}$, what is the minimum magnitude of the force, presumed to be constant, that acts on this neutron?

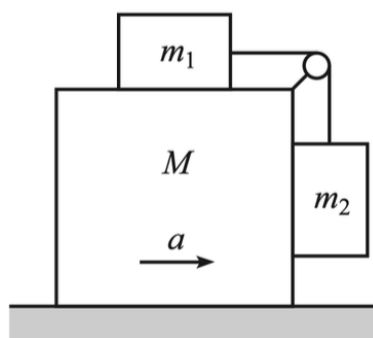
The neutron's mass is $1.67 \times 10^{-27} \text{ kg}$.

3) Consider a pendulum made of a stick of mass M and length L . You hold the pendulum so that it makes the initial angle θ , and let it go. It starts swinging back and forth under the influence of gravity; let T be the period of this motion.

By considering the dimensions of the quantities, try to find the form of T as a function of M , L , θ and g . (i.e. there is a combination of M, L, g, θ so that they together has the dimension of time. This is a candidate for the dependence of T on M , L and g etc).

The answer you just found should involve an undetermined function $f(\theta)$. What can you say about this function based on symmetry?

4) a) All of the surfaces in the setup below are frictionless. You push on the large block and give it an acceleration a . For what value of a is there no relative motion among the masses?



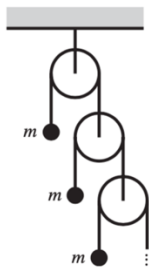
b) Now assume that the string is not massless. It has mass per unit length ρ . Assume that the string has total length L and the horizontal segment has length x . How does that change the answer in a)?

5) Consider the N pulleys system. A string passes over each pulley, with one end attached to a mass m and the other end attached to another pulley. For the N -th pulley, the other end of the string is attached to a mass m . All the pulleys and strings are massless. The masses are held fixed and then simultaneously released.

a) Let $N = 2$, calculate the accelerations of all the masses?

b) Let $N = 3$, calculate the accelerations of all the masses?

c) Let N goes to infinity, what is the acceleration of the top mass?



6) (optional)

A spring with spring constant k , natural length L and total mass M is initially at rest.

a) A spring has spring constant k . If it is cut in half, what is the spring constant of each of the resulting shorter springs?

b) Consider pulling the spring at its right end with a force F towards the right. The left end is not fixed to anything and the entire system moves forward. How long would the spring become?